Tanmay Bhuskute

+1 (732) 858-3027 | tanmay.v.bhuskute@gmail.com | github.com/Tanmayb05 | linkedin.com/in/tanmay-bhuskute
Pittsburgh, PA

SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production engineering experience building Python automation, cloud infrastructure, and CI/CD release pipelines on AWS. Skilled in Python, PySpark, GitHub, and application design for scalable, risk-aware software solutions.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, TypeScript, C/C++, SQL, Go.

Data & Automation: PySpark, Delta Lake, pandas, NumPy, scikit-learn, Process Improvement.

Web & Backend: React, Node.js, Django, Flask, FastAPI, REST APIs, Application Design.

Databases: PostgreSQL, MySQL.

Developer Tools: AWS, Git, GitHub, GitLab CI/CD, Jenkins, Release Management, Docker, Kubernetes, ArgoCD, Helm, Terraform, Linux, Bash, SonarQube, Azure, Google Cloud Platform.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Led technical design of a two-stage memory-augmented LLM pipeline for Android GUI bug reproduction using Python, decoupling video understanding from device automation, evaluated across 53 runs on 20+ apps.
- Authored technical documentation and design artifacts benchmarking the system against **ViBR** baseline, demonstrating **25.7x latency reduction** (39.4s vs 1012.1s avg per run).

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Designed and developed **Teamcenter X** infrastructure automation solutions with **Python** (Boto3) on **AWS**, replacing legacy Ruby scripts and reducing deployment time by **40%**.
- Built **CI/CD** release pipelines with **GitLab** and GitHub-integrated workflows, executing **44 production upgrades** in 4 months while diagnosing live infrastructure issues across Linux, EC2, and databases.
- Collaborated with architects and engineers to design an **Ansible-based Blue-Green deployment** solution, reducing production upgrade downtime by **50%** and improving release management reliability.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, and GitLab CI/CD, aligning solutions with enterprise security and risk management standards.
- Served as Quality Coach and SonarQube Specialist, driving automated testing adoption to raise code coverage from **35% to 78%**; authored onboarding documentation and mentored **5 engineers**.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of **AWS** resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** automated testing framework, catching **95%** of misconfigurations pre-production.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [Python, PySpark, Delta Lake, scikit-learn, FastAPI] Jul 2026

- Designed and developed ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** dataset) using **PySpark**; two-stage Poisson hurdle model cut RMSE **43%** over baseline.

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence-grounded **RAG** solution with citation-backed answers, designing user experience for high-stakes healthcare eligibility decisions.
- Designed **17-tool MCP server** exposing structured tool-calling APIs, consumable by external AI agents for cross-payer comparison workflows.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).